Entregable Final - Tecnologia de Base de Datos I

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Introduccion

El presente informe documenta la migracion completa de la base de datos `employees` desde MariaDB 11.8.9 hacia PostgreSQL 18, incluyendo la migracion de tablas (datos), la migracion de vistas y las consultas de verificacion para validar la integridad y consistencia de los datos. La migracion se realiza sobre contenedores Docker en el entorno del laboratorio.

Punto 1: Migracion de Tablas (15 pts)

1.1 Exportacion de datos desde MariaDB

Verificacion de tablas en MariaDB:

```
SHOW TABLES;
```

Salida:

```
+----------------------+
| Tables_in_employees  |
+----------------------+
| current_dept_emp     |
| departments          |
| dept_emp             |
| dept_emp_latest_date |
| dept_manager         |
| employees            |
| salaries             |
| titles               |
+----------------------+
```

Analisis de tamano de tablas:

```
SELECT TABLE_NAME, table_rows, data_length, index_length
FROM information_schema.tables
WHERE table_schema = 'employees'
ORDER BY data_length DESC;
```

Salida:

```
+----------------------+------------+-------------+------
| TABLE_NAME           | table_rows | data_length | index
+----------------------+------------+-------------+------
| salaries             |    2838426 |   100270080 |
| titles               |     441654 |    20512768 |
| employees            |     298759 |    15220736 |
| dept_emp             |     331143 |    12075008 |
| dept_manager         |         24 |       16384 |
| departments          |          9 |       16384 |
| dept_emp_latest_date |       NULL |        NULL |
| current_dept_emp     |       NULL |        NULL |
+----------------------+------------+-------------+------
```

1.2 Importacion en PostgreSQL con pgloader

Creacion de la base destino:

```
psql -h 192.168.56.55 -U brody -d postgres -c "CREATE DAT
```

Archivo migracion.load:

```
LOAD DATABASE
    FROM mysql://root:x@127.0.0.1:3306/employees
    INTO postgresql://brody:x@192.168.56.55:5432/pdb_empl

WITH include drop,
    create tables,
    create indexes,
    reset sequences,
    workers = 8, concurrency = 2,
    batch rows = 10000

SET maintenance_work_mem to '256MB',
    work_mem to '32MB'

CAST type datetime to timestamp drop default drop not nul
       using zero-dates-to-null,
     type enum to text
;
```

Ejecucion:

```
pgloader migracion.load
```

Salida:

```
             table name     errors       rows      bytes
-----------------------  ---------  ---------  ---------
        fetch meta data          0         21
         Create Schemas          0          0
       Create SQL Types          0          0
          Create tables          0         12
         Set Table OIDs          0          6
```

```
----------------------  ---------  ---------  ---------
     employees.salaries          0    2844047    94.2 MB
       employees.titles          0     443308    16.9 MB
    employees.employees          0     300024    13.2 MB
     employees.dept_emp          0     331603    10.7 MB
  employees.departments          0          9     0.1 kB
 employees.dept_manager          0         24     0.8 kB
----------------------  ---------  ---------  ---------
COPY Threads Completion          0          8
         Create Indexes          0          9
 Index Build Completion          0          9
        Reset Sequences          0          0
           Primary Keys          0          6
    Create Foreign Keys          0          6
        Create Triggers          0          0
        Set Search Path          0          1
       Install Comments          0          0
----------------------  ---------  ---------  ---------
      Total import time         OK    3919015   134.9 MB
```

1.3 Verificacion de datos cargados

Conteo de filas en MariaDB:

```
SELECT 'departments' AS tabla, COUNT(*) AS filas FROM dep
UNION ALL SELECT 'dept_emp', COUNT(*) FROM dept_emp
UNION ALL SELECT 'dept_manager', COUNT(*) FROM dept_manag
UNION ALL SELECT 'employees', COUNT(*) FROM employees
UNION ALL SELECT 'salaries', COUNT(*) FROM salaries
UNION ALL SELECT 'titles', COUNT(*) FROM titles;
```

Salida:

```
+--------------+---------+
| tabla        | filas   |
+--------------+---------+
| departments  |       9 |
```

```
| dept_emp     |  331603 |
| dept_manager |      24 |
| employees    |  300024 |
| salaries     | 2844047 |
| titles       |  443308 |
+--------------+---------+
```

Conteo de filas en PostgreSQL:

```
SELECT 'departments' AS tabla, COUNT(*) AS filas FROM dep
UNION ALL SELECT 'dept_emp', COUNT(*) FROM dept_emp
UNION ALL SELECT 'dept_manager', COUNT(*) FROM dept_manag
UNION ALL SELECT 'employees', COUNT(*) FROM employees
UNION ALL SELECT 'salaries', COUNT(*) FROM salaries
UNION ALL SELECT 'titles', COUNT(*) FROM titles;
```

Salida:

```
    tabla     |  filas
--------------+---------
 departments  |       9
 dept_emp     |  331603
 dept_manager |      24
 employees    |  300024
 salaries     | 2844047
 titles       |  443308
(6 filas)
```

Resultado: Los conteos coinciden exactamente en las 6 tablas.

Punto 2: Migracion de Vistas (10 pts)

2.1 Extraccion de definiciones desde MariaDB

Listado de vistas:

```
SELECT TABLE_NAME FROM information_schema.views WHERE tab
```

Salida:

```
+----------------------+
| TABLE_NAME           |
+----------------------+
| current_dept_emp     |
| dept_emp_latest_date |
+----------------------+
```

Definicion original de la vista dept_emp_latest_date:

```
SHOW CREATE VIEW dept_emp_latest_date\G
```

Salida:

```
CREATE ALGORITHM=UNDEFINED DEFINER=`root`@`%` SQL SECURIT
select `dept_emp`.`emp_no` AS `emp_no`, max(`dept_emp`.`f
max(`dept_emp`.`to_date`) AS `to_date` from `dept_emp` gr
```

Definicion original de la vista current_dept_emp:

```
SHOW CREATE VIEW current_dept_emp\G
```

Salida:

```
CREATE ALGORITHM=UNDEFINED DEFINER=`root`@`%` SQL SECURIT
select `l`.`emp_no` AS `emp_no`, `d`.`dept_no` AS `dept_n
`l`.`to_date` AS `to_date` from (`dept_emp` `d` join `dep
on(`d`.`emp_no` = `l`.`emp_no` and `d`.`from_date` = `l`.
```

2.2 Adaptacion de sintaxis a PostgreSQL

Adaptaciones realizadas:

MariaDB	PostgreSQL	Motivo
`ALGORITHM=UNDEFINED`	(eliminado)	Especifico de MariaDB
`DEFINER=root@%`	(eliminado)	Especifico de MariaDB
`SQL SECURITY DEFINER`	(eliminado)	Especifico de MariaDB
Backticks `	(eliminado)	PostgreSQL usa comillas dobles
`max()`	`MAX()`	Misma funcion
`group by`	`GROUP BY`	Misma clausula

2.3 Creacion de vistas en PostgreSQL

Vista 1: dept_emp_latest_date

```
CREATE OR REPLACE VIEW dept_emp_latest_date AS
SELECT
    emp_no,
    MAX(from_date) AS from_date,
    MAX(to_date) AS to_date
FROM dept_emp
GROUP BY emp_no;
```

Salida: `CREATE VIEW`

Vista 2: current_dept_emp

```
CREATE OR REPLACE VIEW current_dept_emp AS
SELECT
    l.emp_no,
    d.dept_no,
    l.from_date,
    l.to_date
FROM dept_emp d
```

```
JOIN dept_emp_latest_date l
    ON d.emp_no = l.emp_no
    AND d.from_date = l.from_date
    AND l.to_date = d.to_date;
```

Salida: `CREATE VIEW`

Listado de vistas en PostgreSQL:

```
\dv
```

Salida:

```
               Listado de vistas
 Esquema |        Nombre         | Tipo  | Dueno
---------+-----------------------+-------+-------
 public  | current_dept_emp      | vista | brody
 public  | dept_emp_latest_date | vista | brody
(2 filas)
```

Prueba de las vistas:

```
SELECT * FROM dept_emp_latest_date LIMIT 5;
```

Salida:

```
 emp_no | from_date  |  to_date
--------+------------+------------
  10001 | 1986-06-26 | 9999-01-01
  10002 | 1996-08-03 | 9999-01-01
  10003 | 1995-12-03 | 9999-01-01
  10004 | 1986-12-01 | 9999-01-01
  10005 | 1989-09-12 | 9999-01-01
(5 filas)

SELECT * FROM current_dept_emp LIMIT 5;
```

Salida:

```
 emp_no | dept_no | from_date  |  to_date
--------+---------+------------+------------
  10004 | d004    | 1986-12-01 | 9999-01-01
  10286 | d009    | 1994-12-24 | 1999-01-24
  10287 | d005    | 1997-09-05 | 9999-01-01
  10293 | d004    | 1992-12-12 | 9999-01-01
  10295 | d002    | 1995-06-19 | 9999-01-01
(5 filas)
```

Conteo de filas en las vistas:

```
SELECT 'dept_emp_latest_date' AS vista, COUNT(*) FROM dep
UNION ALL SELECT 'current_dept_emp', COUNT(*) FROM curren
```

Salida:

```
        vista          | count
-----------------------+--------
 dept_emp_latest_date | 300024
 current_dept_emp     | 300024
(2 filas)
```

Punto 3: Consultas de Verificacion (25 pts)

3.1 Conteo de filas

Verificado en Punto 1.3: Los conteos coinciden en las 6 tablas.

3.2 Checksums MD5

Checksum de departments en MariaDB:

```
SELECT MD5(GROUP_CONCAT(dept_no, dept_name ORDER BY dept_
```

Salida:

```
+----------------------------------+
| checksum_departments             |
+----------------------------------+
| 50e4426051fe848a64e165300f0ce558 |
+----------------------------------+
```

Checksum de departments en PostgreSQL:

```
SELECT MD5(STRING_AGG(dept_no || dept_name, ',' ORDER BY
```

Salida:

```
       checksum_departments
----------------------------------
 50e4426051fe848a64e165300f0ce558
(1 fila)
```

Resultado: Los checksums coinciden exactamente. La integridad de los datos esta verificada.

3.3 Integridad referencial (huerfanos)

Verificacion 1: dept_emp sin empleado valido

```
SELECT COUNT(*) AS huerfanos FROM dept_emp de LEFT JOIN e
```

Salida: `0`

Verificacion 2: dept_emp sin departamento valido

```
SELECT COUNT(*) AS huerfanos FROM dept_emp de LEFT JOIN d
```

Salida: `0`

Verificacion 3: salaries sin empleado valido

```
SELECT COUNT(*) AS huerfanos FROM salaries s LEFT JOIN em
```

Salida: `0`

Verificacion 4: titles sin empleado valido

```
SELECT COUNT(*) AS huerfanos FROM titles t LEFT JOIN empl
```

Salida: `0`

Verificacion 5: dept_manager sin empleado valido

```
SELECT COUNT(*) AS huerfanos FROM dept_manager dm LEFT JO
```

Salida: `0`

Verificacion 6: dept_manager sin departamento valido

```
SELECT COUNT(*) AS huerfanos FROM dept_manager dm LEFT JO
```

Salida: `0`

Resultado: No hay registros huerfanos. La integridad referencial esta garantizada.

3.4 Analisis de resultados

Verificacion	MariaDB	Postgre
Conteo de filas	3,919,015	3,919,015
Checksum departments	50e4426051fe848a64e165300f0ce558	50e4426051fe848a64

Huerfanos	0	0
Vistas	2	2

Conclusion: La migracion fue 100% exitosa. Todos los datos, vistas e integridad referencial se preservaron correctamente.

Backup de PostgreSQL

Comando utilizado:

```
pg_dump -h 192.168.56.55 -U brody -Fc -f ~/entregable_fin
```

Resultado:

```
-rw-r--r-- 1 brody brody 34M sep 28 15:46 pdb_employees_2
-rw-r--r-- 1 brody brody 135M sep 28 15:46 pdb_employees_
```

Verificacion del backup:

```
pg_restore -l ~/entregable_final/pdb_employees_20260928.d
```

Salida:

```
; Archive created at 2026-09-28 15:46:01 -04
;     dbname: pdb_employees
;     TOC Entries: 34
;     Compression: gzip
;     Format: CUSTOM
;     Dumped from database version: 18.6 (Debian 18.6-1.p
```

Contenido del backup:

- 6 tablas: departments, dept_emp, dept_manager, employees, salaries, titles

- 6 TABLE DATA (los datos de cada tabla)
- 2 vistas: dept_emp_latest_date, current_dept_emp
- Constraints: PRIMARY KEY, FOREIGN KEYs

Conclusiones

- Se migraron exitosamente las 6 tablas de la base employees desde MariaDB 11.8.9 hacia PostgreSQL 18.
- Se migraron 3,919,015 filas en 31.3 segundos sin errores.
- Se migraron las 2 vistas (dept_emp_latest_date y current_dept_emp) adaptando la sintaxis correctamente.
- Se verifico la integridad de los datos mediante conteos, checksums MD5 e integridad referencial.
- Los checksums coinciden exactamente entre ambos SGBD.
- No existen registros huerfanos en ninguna tabla.
- Se genero un backup completo de PostgreSQL en formato custom y SQL con fecha.

Referencias

- PostgreSQL Global Development Group. (2026). PostgreSQL 18 Documentation. https://www.postgresql.org/docs/
- pgloader Documentation. (2026). pgloader Reference Manual. https://pgloader.readthedocs.io/
- MariaDB Foundation. (2026). MariaDB Documentation. https://mariadb.com/kb/en/documentation/
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